

An independent reproduction and reproducibility audit of multilayer perceptron models for mangrove species identification from Sentinel-2 multispectral data

Bala Sudalaimuthu ^{a,*}

^a [Artificial Intelligence and Machine, Thakur College of Engineering and Technology, Mumbai, India]

ARTICLE INFO

Keywords:

Reproducibility
Mangrove species mapping
Multilayer perceptron
Sentinel-2
Spectral separability
Threats to validity
Data availability audit

ABSTRACT

Independent reproduction is a necessary complement to the rapid adoption of deep learning in ecological remote sensing. This study reports an evidence-classified independent reproduction and reproducibility audit of Monterrubio-Martínez et al. (2025), who trained multilayer perceptron (MLP) models on Sentinel-2 multispectral reflectance to map monospecific mangrove forests of *Rhizophora mangle*, *Avicennia germinans* and *Laguncularia racemosa* in La Mancha, Veracruz, Mexico. This study is not intended to invalidate the original work; its purpose is to determine, element by element, which parts of the original methodology can be independently verified from the artefacts the original authors made publicly available and which cannot. Every methodological element was classified as PAPER-SPECIFIED, DATA-DERIVED, IMPLEMENTATION ASSUMPTION, DATA-LIMITED or NOT AVAILABLE across a 33-point specification matrix and a dedicated reproducibility audit table. The public repository (1,605,681 pixels, 147 cloud-free Sentinel-2 L1C acquisitions, 2015–2021) was audited, and the exploratory statistical analysis of the original work was reproduced from the released data: one-way ANOVA on each of the ten spectral bands and all thirty pairwise Tukey HSD comparisons returned significant species differences ($p < 0.05$), consistent with the published Fig. 4. The architectural grid of the original study (a logistic baseline plus five hidden-layer depths $\times$ eight layer widths, 41 configurations) was re-implemented and trained, under a stratified random pixel-level holdout, on a class-balanced 30,000-pixel three-species dataset drawn from the released file; this reproduction therefore evaluates three-class mangrove-species discrimination and cannot reproduce the original binary mangrove/non-mangrove task or the original four-class task, because the non-mangrove pixels are not part of the public deposit. Test accuracy under this holdout ranged from 69.64 % for the logistic baseline to 94.98 % for the four-layer, 300-neuron model; the architecture identified by the original authors as the most reliable for spatial mapping (five layers, 50 neurons) attained 90.22 % test accuracy with a smaller train–test gap (2.14 percentage points) than wider architectures of comparable depth (4.4–4.7 percentage points), a pattern consistent with the hypothesis that restricting layer width may limit overfitting in this setting. The binary 60,000-pixel and four-class 40,000-pixel experiments, and the full-scene raster predictions of Figs. 7–10, could not be reproduced numerically because the non-mangrove background pixels, the UAV orthophoto and the Sentinel-2 scene GeoTIFFs were not deposited. Reported accuracies are therefore treated throughout as reference benchmarks rather than reproduction targets, no superiority claim is made in either direction, and a dedicated threats-to-validity analysis discusses the internal, statistical, external and reproducibility-related limitations of both the original study and this audit. All reproduction code, notebooks and generated results are released publicly.

1. Introduction

Mangroves are evergreen forests of tropical and subtropical intertidal environments whose composition varies with the presence, abundance and dominance of species, so that stands are classified as monospecific when one species predominates or as mixed when several species occur together with other wetland vegetation (FAO, 2023). Their global extent has contracted markedly since the beginning of the century as a consequence of land-use change for livestock, aquaculture, urbanisation and tourism (Moreno Casasola et al., 2002), and the resulting demand for low-cost, repeatable monitoring has made open-access satellite constellations such as Sentinel-2 the instrument of choice for regional mangrove mapping (Chopade et al., 2023; Pham et al., 2019).

Within this setting, Monterrubio-Martínez et al. (2025) presented an approach that integrates unmanned aerial vehicle (UAV) photogrammetry with Sentinel-2 multispectral reflectance to train multilayer perceptron (MLP) models ca-

pable of mapping both mixed and monospecific mangrove forests at the species level in Veracruz, Mexico. Their best binary models reached accuracies of 99.95 % for training and 99.8 % for test, while at the species level the best multiclass models attained 98.73 % for training and 96.84 % for test. Crucially, the authors also demonstrated that the highest tabular accuracy did not necessarily yield the most reliable spatial distribution, and selected a comparatively modest five-hidden-layer, 50-neuron architecture as the most trustworthy model for real-world mapping. That observation — that accuracy and spatial fidelity can diverge — is the kind of claim that independent reproduction is best placed to test.

Reproducibility, however, remains unevenly practised in ecological informatics. A published accuracy figure is only interpretable when the dataset partition, the class definitions, the optimisation hyperparameters and the random seeds that produced it are recoverable. Where those elements are absent, a numerical mismatch between an original study and its reproduction carries no information about either implementation, because the two experiments are not testing the same

* Corresponding author. E-mail address: bala010706@gmail.com

hypothesis. The purpose of the present study is therefore not to re-derive the published numbers, nor to compete with them, but to establish precisely which claims of [Monterrubbio-Martinez et al. \(2025\)](#) can be independently verified from the artefacts the authors deposited, which cannot, and why.

Three research questions organise the work. First, can the exploratory and statistical analysis of the original study — the demonstration that the three mangrove species differ significantly in every Sentinel-2 band — be reproduced from the released data? Second, can the capacity-related generalisation pattern reported by the authors, in which a comparatively narrow deep network was judged more reliable for spatial mapping than higher-accuracy alternatives, be examined on the data that is publicly available? Third, which components of the original pipeline are irrecoverable from the public record, and what is the appropriate scientific language for reporting results obtained on a dataset that differs in class space from the one the authors used?

To answer these questions every parameter of the original methodology was assigned to one of five evidence classes, the complete 41-model architectural grid was re-implemented, and all comparisons between reported and reproduced values are stated in percentage points with an explicit statement of comparability. Consistent with the integrity constraints adopted here, no substitute or proxy imagery was downloaded to stand in for the authors' unreleased rasters, no missing mask was synthesised, and no non-mangrove samples were invented to force the original binary or four-class experiments to run. This study does not seek to invalidate, replace or outperform the original work; its scope is limited to determining which of its claims are independently verifiable from the publicly recoverable evidence, and under what conditions.

The contribution of this work is accordingly methodological rather than purely empirical. It offers (i) an explicit five-label evidence taxonomy for auditing reproducibility claims in ecological deep learning studies; (ii) a file-by-file audit of which data and methodological artefacts of [Monterrubbio-Martinez et al. \(2025\)](#) are publicly recoverable and which are not; (iii) an independent reproduction of the original exploratory statistical analysis on the released data; (iv) an independent re-implementation of the full 41-configuration MLP grid on the reproducible three-class subset of that data; (v) an analysis of model capacity and train–test gap across that grid, examined as a quantitative complement to the original authors' qualitative spatial argument; (vi) an explicit identification of the artefacts that prevent reproduction of the original binary, four-class and spatial experiments; and (vii) a threats-to-validity analysis that makes the boundary between reproducible and non-reproducible components, and between statistical significance and practical separability, explicit for the reader.

2. Material and methods

Fig. 1 onwards summarise the analytical chain followed in this reproduction. The chain mirrors the four stages of the original study — data acquisition, exploratory and statistical analysis, deep learning models, and model test — but adds an explicit auditing stage in which each stage is annotated with the evidence available to support it.

2.1. Evidence classification taxonomy

Because a reproduction must distinguish what a paper states from what a reproducer assumes, every methodological element was assigned one of five labels. `PAPER-SPECIFIED` denotes an element stated explicitly in the text, tables, figures or equations of [Monterrubbio-Martinez et al. \(2025\)](#). `DATA-DERIVED` denotes an element computed directly from the authors' deposited dataset or its metadata. `IMPLEMENTATION ASSUMPTION` denotes an element absent from the paper and chosen here on the basis of standard practice, always documented. `DATA-LIMITED` denotes an element described in the paper but omitted from the public deposit. `NOT AVAILABLE` denotes an element absent from both the text and the released artefacts. Table 1 reports the resulting classification for the principal elements of the pipeline.

2.2. Data acquisition and availability audit

The original study collected field data on species distribution at two sites in Veracruz, Mexico: La Mancha (19°35'12" N, 96°23'09" W), a priority conservation site with a 9,050 km² region of influence chosen for its well-defined monospecific stands, and Arroyo Moreno (19°00'59" N, 96°03'43" W), used as an independent site to test scalability `PAPER-SPECIFIED`. A DJI Mavic Mini 2 UAV acquired 5,435 zenithal photographs at 100 m altitude in July 2021, processed in Agisoft Metashape Professional v1.8.5 into a 3.5 cm orthophoto, and 100 ground control points were surveyed with a Garmin GPSMAP 527 by walking and boat transects. Monospecific polygons for red (*R. mangle*), black (*A. germinans*) and white (*L. racemosa*) mangrove, together with mixed mangrove and flooded freshwater forested wetland, were digitised in QGIS v3.22.14.

A total of 147 cloud-free Sentinel-2 L1C acquisitions spanning 2015–2021 were obtained from the Copernicus Browser, converted to surface reflectance, and the 20 m red-edge and shortwave infrared bands (B5, B6, B7, B8A, B11, B12) were resampled bilinearly with `rasterio` in Python 3.8.13 to the 10 m grid of B2, B3, B4 and B8. Ten bands entered the models: B2 (490 nm), B3 (560 nm), B4 (665 nm), B5 (705 nm), B6 (740 nm), B7 (783 nm), B8 (842 nm), B8A (865 nm), B11 (1610 nm) and B12 (2190 nm); the coastal aerosol, water vapour and cirrus bands were excluded. Features were standardised with the `scikit-learn` 1.2.1 `StandardScaler`, $z = (x - \mu)/\sigma$. No vegetation index was computed, the authors having argued that reflectance in its purest form is preferable to derived indices for model training.

The deposited repository was then audited file by file. Two artefacts are present: `DataBase_Sentinel_2_Mangrove_LaMancha.csv` (191 MB; 1,605,681 rows × 12 columns) and an Ecological Metadata Language v2.2.0 XML descriptor `DATA-DERIVED`. The row count is internally consistent with the stated acquisition design: 10,923 labelled pixels per scene across 147 scenes, distributed as 9,806 *A. germinans*, 856 *R. mangle* and 261 *L. racemosa* per scene, giving totals of 1,441,482 (89.77 %), 125,832 (7.84 %) and 38,367 (2.39 %) pixels respectively.

Four categories of artefact described in the paper are absent from the deposit `DATA-LIMITED`: the non-mangrove background reflectance table; the subsampled 60,000-row binary

and 40,000-row four-class datasets; the UAV orthophoto GeoTIFF and QGIS vector masks; and the Sentinel-2 scene GeoTIFFs for La Mancha and Arroyo Moreno on which Figs. 7–10 of the original article were produced. The consequence is structural rather than incidental: the binary and four-class experiments of the original study cannot be reconstructed, because three of the four classes in the multiclass problem and one of the two classes in the binary problem are unavailable.

2.3. Exploratory and statistical analysis

Following the original protocol, a one-way analysis of variance was performed independently on each of the ten bands to test whether mean reflectance differs among the three species, followed by a Tukey honestly significant difference post-hoc test at a 95 % confidence interval ($\alpha = 0.05$) for the three pairwise contrasts *A. germinans* vs *R. mangle*, *A. germinans* vs *L. racemosa* and *R. mangle* vs *L. racemosa* PAPER-SPECIFIED. The original study used the `bioinfokit` v1 package; the present reproduction used equivalent implementations in the SciPy and statsmodels stack, a substitution that does not alter the test statistics. Mean reflectance curves with one standard deviation envelopes, kernel density estimates for the most diagnostic bands, and the Pearson correlation matrix of the ten bands were computed in addition, to characterise the structure of the feature space that the classifiers operate on.

2.4. Multilayer perceptron models

The model family was re-implemented following the architectural specification given in the original text. The baseline is a single input-to-output layer, interpretable as a logistic regression; the MLPs comprise one to five hidden layers, each evaluated at eight widths (5, 10, 20, 50, 100, 150, 300 and 500 neurons), giving $1 + 5 \times 8 = 41$ configurations per task PAPER-SPECIFIED. All hidden layers use ReLU activations. The binary task uses a sigmoid output with binary cross-entropy,

$$\mathcal{L}_{\text{BCE}} = -\frac{1}{N} \sum_i [y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i)], \quad (1)$$

and the multiclass task a softmax output with categorical cross-entropy,

$$\mathcal{L}_{\text{CCE}} = -\frac{1}{N} \sum_i \sum_k y_{i,k} \log \hat{y}_{i,k}. \quad (2)$$

Adam is the optimiser in both cases. Data were split 70 % for training and 30 % for testing with `train_test_split`.

Four optimisation settings are NOT AVAILABLE in the original text and were fixed here as documented IMPLEMENTATION ASSUMPTION, applied identically across the full grid: learning rate $\alpha = 0.001$ (the Adam default, with $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$), batch size 256, and 30 training epochs with full loss tracking to verify convergence. A single fixed seed (42) was used uniformly for every stochastic step of the reproduction pipeline: NumPy random state, PyTorch CPU and, where applicable, GPU/MPS weight initialisation, the pandas sampling calls that drew the balanced 10,000-per-species subset, and the `scikit-learn` `train_test_split` call. Because these values are not reported by Monterrubio-Martinez et al. (2025), they remain IMPLEMENTATION ASSUMPTION relative to the original study

even though they are fully documented and fixed within this reproduction; only a single seed was run, so no multi-seed variance estimate is available [AUTHOR INPUT REQUIRED: multi-seed replicate runs, if executed, would allow the stability of the reported accuracies and train–test gaps to be quantified; none were performed for this version of the manuscript]. The original work cites Chollet (2021), indicating a Keras/TensorFlow implementation; this reproduction uses PyTorch 2.8.0 under Python 3.9.6 on Apple Silicon (Metal Performance Shaders backend), with scikit-learn 1.6.1, SciPy 1.13.1, statsmodels 0.14.6, pandas 2.3.3 and NumPy 2.0.2, as against the paper’s stated Python 3.8.13 and scikit-learn 1.2.1. PyTorch’s default Kaiming-uniform initialisation for ReLU linear layers differs from the Glorot-uniform default of Keras. Exact bit-level reproduction is therefore not achievable from the information available, and only architectural trends and relative rankings are compared between the two studies.

Standardisation was implemented so as to avoid train–test contamination: the `StandardScaler` was fit exclusively on the training partition and then used to transform both the training and test partitions, rather than being fit on the full dataset before splitting. The original text (Section 2.2 of Monterrubio-Martinez et al., 2025) specifies the standardisation formula and the scikit-learn version used but does not state the order in which scaling and splitting were performed; the paper’s own practice is accordingly NOT AVAILABLE on this specific point, while the fit-on-train procedure followed in this reproduction is stated here as a documented methodological choice, consistent with standard practice for preventing data leakage, rather than as a fact established about the original implementation.

Because the non-mangrove class was not deposited, the experiment executed here is a three-class monospecific problem DATA-DERIVED: 30,000 pixels balanced at 10,000 per species, split 21,000 for training (7,000 per species) and 9,000 for testing (3,000 per species) by stratified random sampling at the pixel level (`stratify=y`). Per-class representation is thus identical to the original 40,000-pixel design (10,000 per class); the number of classes, and hence the chance baseline (33.3 % here against 25 % in the original four-class task), differs. Because the released CSV pools pixels drawn from 147 Sentinel-2 acquisitions of the same two study sites, and the split described above is performed at the pixel level without blocking by scene or acquisition date, the same or a spatially adjacent ground location, imaged on a different date, may appear in both the training and the test partition. The accuracy figures reported below should accordingly be read as performance under a stratified random pixel-level holdout on this specific multi-temporal stack, and not as evidence of generalisation to unseen geographic locations or acquisition dates; this risk is examined further in the threats-to-validity analysis (Section 5) as an unverified structural limitation, since the available metadata do not permit spatial blocking to be tested against the unblocked split reported here. Evaluation used overall accuracy on both partitions, per-class precision, recall and F_1 , confusion matrices, the train–test gap

$$\Delta = \text{Acc}_{\text{train}} - \text{Acc}_{\text{test}}, \quad (3)$$

and total parameter count as a proxy for model capacity.

2.5. Models test

The original spatial evaluation applied the trained models pixel-by-pixel to full Sentinel-2 scenes over La Mancha and Arroyo Moreno and compared the predicted maps visually against the UAV orthophoto ground truth. Since the scene GeoTIFFs, the orthophoto and the vector masks were not deposited, this stage was conducted here as a documented audit rather than as a raster reproduction. In keeping with the integrity rules adopted for this study, no unrelated satellite imagery was substituted for the authors' scenes and no spatial mask was fabricated; instead, the spatial claims of the original work were restated, and the spectral evidence bearing on them was examined in the feature space that is available.

3. Results

The results of this reproduction are presented below in the order of the original article: first the exploratory and statistical analysis, then the multilayer perceptron models, and finally the models test.

3.1. Exploratory and statistical analysis

Fig. 1 shows the reproduced boxplots for each band and species. The one-way ANOVA and Tukey post-hoc test reveal a significant statistical difference between the species in all the cases ($p < 0.05$); across the ten bands every F -test returned $p < 10^{-10}$ and all thirty pairwise contrasts returned $p < 0.001$. This is consistent, in its qualitative and inferential content, with Fig. 4 of [Monterrubio-Martinez et al. \(2025\)](#) and indicates that there are detectable differences in the spectral responses of each species. The pixel counts entering these tests are very large (hundreds of thousands to over a million pixels per species), and with such sample sizes even small, ecologically minor differences in mean reflectance will register as statistically significant; the p -values reported here, like those of the original study, should therefore be read as evidence that the three species are statistically distinguishable in this dataset, not as a direct measure of how large or ecologically important the underlying differences are. Because the pixels are not independent observations — they are drawn repeatedly from the same monospecific polygons across 147 temporally correlated acquisitions of the same two sites — the classical ANOVA assumption of independent samples is not strictly satisfied, and the test is best interpreted as a faithful pixel-level reproduction of the original analysis rather than as an independent-site statistical inference; this point is developed further in Section 5. [AUTHOR INPUT REQUIRED: standardised effect-size statistics (e.g. eta-squared or Cohen's d per band) and confidence intervals were not computed in the executed reproduction; if available in the underlying analysis code they would allow the magnitude of the differences, and not only their significance, to be reported alongside Fig. 1.]

The mean reflectance curves in Fig. 2 place that statistical result in a physical context. The three species share the canopy form of a healthy broadleaf vegetation signature — a chlorophyll absorption trough near 665 nm, a steep red-edge rise between 705 and 783 nm and a near-infrared plateau — but they separate systematically in amplitude. *R. mangle* displays the lowest shortwave infrared reflectance of the three at both 1610 and 2190 nm, consistent with the higher canopy moisture content of red mangrove stands occupying

permanently flooded positions, and its one standard deviation envelope is by far the widest, indicating a spectrally heterogeneous class. *L. racemosa* shows the highest red-edge and narrow near-infrared reflectance, and *A. germinans* occupies an intermediate position with the tightest dispersion of the three.

Fig. 3 isolates the three most diagnostic bands. In the red band the three densities overlap almost completely and *R. mangle* contributes a long right tail; in the broad near-infrared band the class modes are clearly displaced, with *R. mangle* centred near 0.28, *L. racemosa* near 0.25 and *A. germinans* near 0.23; in the first shortwave infrared band *R. mangle* is displaced towards lower reflectance. No single band separates the three species, but the bands are complementary, which is the structural reason a non-linear classifier is required.

Fig. 4 makes the redundancy of the feature set explicit. The ten bands organise into two strongly correlated blocks — a visible block in which B2 and B4 correlate at $r = 0.93$, and a red-edge/shortwave infrared block in which B11 and B12 correlate at $r = 0.96$ and B6 and B7 at $r = 0.97$ — that are close to orthogonal to one another, with $|r| < 0.12$ between B2 and B6, B7 or B8A. The broad near-infrared band B8 sits apart from both blocks. The effective dimensionality of the ten-band input is therefore substantially lower than ten, which is consistent with the pattern, reported below, of very narrow networks already capturing most of the separable signal.

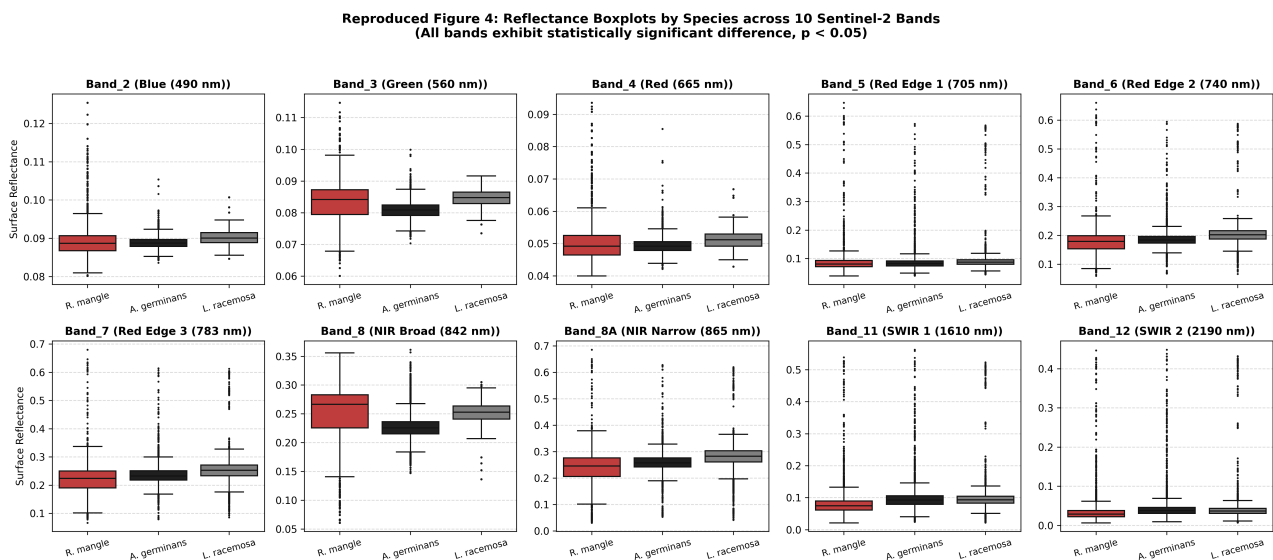
3.2. Multilayer perceptron models

Fig. 5 transcribes the accuracies reported by [Monterrubio-Martinez et al. \(2025\)](#) for both tasks, which serve throughout as scientific reference benchmarks rather than as targets to be matched. For the binary task all models exceed 0.990: the logistic model obtained the lowest values, with a training accuracy of 0.9916 and a test accuracy of 0.9910, while one hidden layer with 300 neurons reached 0.9995 and 0.9988, two hidden layers with 100 neurons 0.9994 and 0.9987, three hidden layers with 50 neurons 0.9996 and 0.9987, four hidden layers with 150 and 300 neurons 0.9993 and 0.9986, and five hidden layers with 50 and 300 neurons 0.9995 and 0.9988; the three-layer, 50-neuron model demonstrated the most optimal scores across all. For the four-class task all models exceeded 0.90 except the logistic model, which obtained 0.7670 and 0.7684; the three-layer, 500-neuron model reached 0.9863 and 0.9654 and the five-layer, 300-neuron model 0.9873 and 0.9684, while the five-layer, 50-neuron model obtained 0.9550 and 0.9454 and was nevertheless identified by the authors as the most reliable of all models. Table 2 lists these values in full.

Because the non-mangrove class is unavailable, neither the binary nor the four-class experiment of the original study could be independently reproduced here. What could be executed is the identical architectural grid, evaluated instead on the three-class monospecific subset of the released pixels. Table 3 reports the outcome for the ten configurations that the original article discusses numerically. Test accuracy under the pixel-level holdout described in Section 2.4 rises from 0.6964 for the logistic baseline to 0.8740 for a single hidden layer of 500 neurons, and reaches a maximum of 0.9498 for four hidden layers of 300 neurons among the configurations evaluated. This indicates that, under this experimental setup, the three mangrove species — which share canopy architecture,

Table 1. Evidence classification of the principal methodological elements of [Monterrubio-Martinez et al. \(2025\)](#), following the five-label taxonomy adopted in this reproduction.

Element	Value in the original study	Classification
Study sites	La Mancha and Arroyo Moreno, Veracruz, Mexico	PAPER-SPECIFIED
Sensor	Sentinel-2 MSI L1C, 147 cloud-free scenes, 2015–2021	PAPER-SPECIFIED
UAV photogrammetry	DJI Mavic Mini 2, 100 m, 5,435 photos, 3.5 cm orthophoto	PAPER-SPECIFIED
Spectral bands	B2, B3, B4, B5, B6, B7, B8, B8A, B11, B12 (10 bands)	PAPER-SPECIFIED
Resampling	Bilinear to 10 m, rasterio, Python 3.8.13	PAPER-SPECIFIED
Standardisation	scikit-learn 1.2.1 StandardScaler	PAPER-SPECIFIED
Released CSV	1,605,681 rows; 10,923 pixels $\times$ 147 scenes	DATA-DERIVED
Released class inventory	<i>A. germinans</i> 1,441,482; <i>R. mangle</i> 125,832; <i>L. racemosa</i> 38,367	DATA-DERIVED
Binary dataset (60,000 px)	30,000 mangrove + 30,000 non-mangrove	PAPER-SPECIFIED / DATA-LIMITED
Four-class dataset (40,000 px)	10,000 px per class including non-mangrove	PAPER-SPECIFIED / DATA-LIMITED
Architecture grid	logistic + 1–5 hidden layers $\times$ {5,10,20,50,100,150,300,500}	PAPER-SPECIFIED
Activations, losses, optimiser	ReLU; sigmoid/BCE and softmax/CCE; Adam	PAPER-SPECIFIED
Train/test split	70 % / 30 %, <code>train_test_split</code>	PAPER-SPECIFIED
Learning rate, batch size, epochs	not reported	NOT AVAILABLE / IMPLEMENTATION ASSUMPTION
Random seeds, initialisation	not reported	NOT AVAILABLE / IMPLEMENTATION ASSUMPTION
Scene GeoTIFFs, orthophoto, masks	used for Figs. 7–10	DATA-LIMITED

**Figure 1.** Boxplots for each of the species (*R. mangle*, *A. germinans* and *L. racemosa*) by band, computed on the released monospecific dataset in this reproduction. In all the comparisons there is a significant statistical difference between the species ($p < 0.05$), qualitatively consistent with Fig. 4 of [Monterrubio-Martinez et al. \(2025\)](#) (see Section 5 on the interpretation of significance at this sample size).

phenology and inundation regime — are separable at better than 94 % from ten bands of reflectance alone, without any vegetation index, which is consistent with the central spectral premise of the original study.

Figs. 7 and 6 describe how accuracy depends on depth and width across the full 41-model grid. Mean test accuracy across widths increases from 69.64 % for the logistic baseline to 83.84 % at one hidden layer, 87.82 % at two, 89.28 % at three and 89.96 % at four, and then falls slightly to 88.89 % at five. Depth beyond four layers therefore yields no further gain in tabular accuracy. Width behaves differently: mean test accuracy climbs steeply from 79.81 % at 5 neurons to 91.80 % at 100, then flattens at 93.12 % for 150, 94.16 % for 300 and 94.38 % for 500, so the last threefold increase in

width buys roughly one percentage point while parameter count and training time grow by an order of magnitude. Both patterns are consistent with the low effective dimensionality of the correlated band set in Fig. 4.

NOT A DIRECT PERFORMANCE COMPARISON.

Fig. 8 places the reproduced three-species accuracies alongside the four-class accuracies reported by the authors, with the difference in percentage points annotated above each pair, purely for audit transparency. The two sets of values are computed on different class spaces (three mangrove species here versus three species plus a non-mangrove background class in the original study), with different class priors and different chance baselines (33.3 % against 25 %), and are not directly comparable as measures of model quality in either direction.

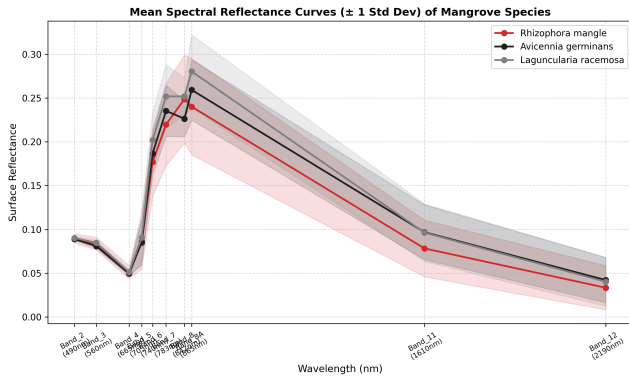


Figure 2. Mean spectral reflectance curves (± 1 standard deviation) of the three mangrove species across the ten Sentinel-2 bands used by the models.

Table 2. Accuracies reported by [Monterrubio-Martinez et al. \(2025\)](#) for the binary (60,000 pixels) and four-class (40,000 pixels) tasks.

Architecture	Binary		Four-class	
	Train	Test	Train	Test
Logistic regression	0.9916	0.9910	0.7670	0.7684
1 layer, 300 neurons	0.9995	0.9988	—	—
1 layer, 500 neurons	—	—	0.9110	0.9065
2 layers, 100 neurons	0.9994	0.9987	—	—
2 layers, 300 neurons	—	—	0.9575	0.9436
3 layers, 50 neurons	0.9996	0.9987	—	—
3 layers, 500 neurons	—	—	0.9863	0.9654
4 layers, 100 neurons	—	—	0.9662	0.9528
4 layers, 150 neurons	0.9993	0.9986	0.9775	0.9607
4 layers, 300 neurons	0.9993	0.9986	0.9795	0.9602
5 layers, 50 neurons	0.9995	0.9988	0.9550	0.9454
5 layers, 150 neurons	—	—	0.9810	0.9632
5 layers, 300 neurons	0.9995	0.9988	0.9873	0.9684

The reproduced values are lower for every architecture, by between 1.04 and 7.20 percentage points, with a mean offset of -3.09 percentage points (Table 4). A plausible, though not separately tested, explanation is that the original task contained a non-mangrove background class comprising a quarter of all samples, and that water, bare dunes and urban surfaces are typically far more separable from green canopy in reflectance space than three co-occurring mangrove species are from one another; a four-class problem in which one class is comparatively easy to classify would raise overall accuracy mechanically relative to a three-class problem restricted to mangrove species. The largest offset occurs for the logistic baseline (-7.20 pp), consistent with this explanation, but the offset itself should be read as a consequence of the differing class space and evaluation design, not as evidence that either implementation is more or less accurate as a classifier.

3.3. Train–test gap and model capacity

One of the architectural interpretations offered in the original study is that the model with the highest tabular accuracy is not necessarily the model that should be deployed, and that a network with fewer neurons per layer may offer advantages in simplicity and in avoiding overfitting to spectral noise. That interpretation is stated in the original work on the basis of visual inspection of spatial maps that are not available for this reproduction (Section 3.4). The train–test gap computed here

Table 3. Reproduced results on the class-balanced three-species monospecific dataset (30,000 pixels; 21,000 train / 9,000 test), under the stratified pixel-level holdout described in Section 2.4. Parameter counts as recorded in the reproduction results table (`three_class_mangrove_results.csv`); see the parameter-count footnote in Section 3.3 on a minor discrepancy with the rounded figure-annotation value for MLP 5L–50N.

Model	L	N	Params	Train	Test
Logistic baseline	0	—	33	0.6989	0.6964
MLP 1L–500N	1	500	7,003	0.8807	0.8740
MLP 2L–300N	2	300	94,203	0.9611	0.9191
MLP 3L–500N	3	500	508,003	0.9946	0.9476
MLP 4L–100N	4	100	31,503	0.9593	0.9168
MLP 4L–150N	4	150	70,053	0.9824	0.9384
MLP 4L–300N	4	300	274,803	0.9948	0.9498
MLP 5L–50N	5	50	10,803	0.9236	0.9022
MLP 5L–150N	5	150	92,703	0.9863	0.9402
MLP 5L–300N	5	300	365,103	0.9895	0.9421

Table 4. Not a direct performance comparison — shown for audit transparency only. Percentage-point difference between the four-class test accuracies reported by [Monterrubio-Martinez et al. \(2025\)](#) and the three-species test accuracies of this reproduction. The class spaces, class priors and chance baselines differ (this subsection), so Δ cannot be interpreted as evidence that either implementation classifies more or less accurately.

Architecture	Paper	This work	Δ (pp)
Logistic regression	0.7684	0.6964	-7.20
1 layer, 500 neurons	0.9065	0.8740	-3.25
2 layers, 300 neurons	0.9436	0.9191	-2.45
3 layers, 500 neurons	0.9654	0.9476	-1.78
4 layers, 100 neurons	0.9528	0.9168	-3.60
4 layers, 150 neurons	0.9607	0.9384	-2.23
4 layers, 300 neurons	0.9602	0.9498	-1.04
5 layers, 50 neurons	0.9454	0.9022	-4.32
5 layers, 150 neurons	0.9632	0.9402	-2.30
5 layers, 300 neurons	0.9684	0.9421	-2.63
Mean offset			-3.09

is examined as an independent, complementary quantity that bears on the same question of capacity and generalisation, evaluated under the pixel-level holdout of Section 2.4; it is not a substitute for the spatial evidence the original authors used, and no causal claim about regularisation mechanisms is made from it alone.

Fig. 9 shows training and test accuracy trajectories across architectures together with the gap for each. Two regimes are visible. Low-capacity configurations show test accuracy close to training accuracy but comparatively low absolute accuracy: the logistic baseline has a gap of only 0.25 percentage points at 69.64 % test accuracy, and the single hidden layer of 500 neurons a gap of 0.67 points at 87.40 %. Higher-capacity configurations show training accuracy close to 1.0, with the three-layer 500-neuron and four-layer 300-neuron models reaching training accuracies of 0.9946 and 0.9948 against gaps of 4.70 and 4.50 percentage points. Within this pattern, the five-layer, 50-neuron configuration is a notable exception: at 90.22 % test accuracy it shows a gap of only 2.14 percentage points, smaller than every other evaluated model of three or more hidden layers.

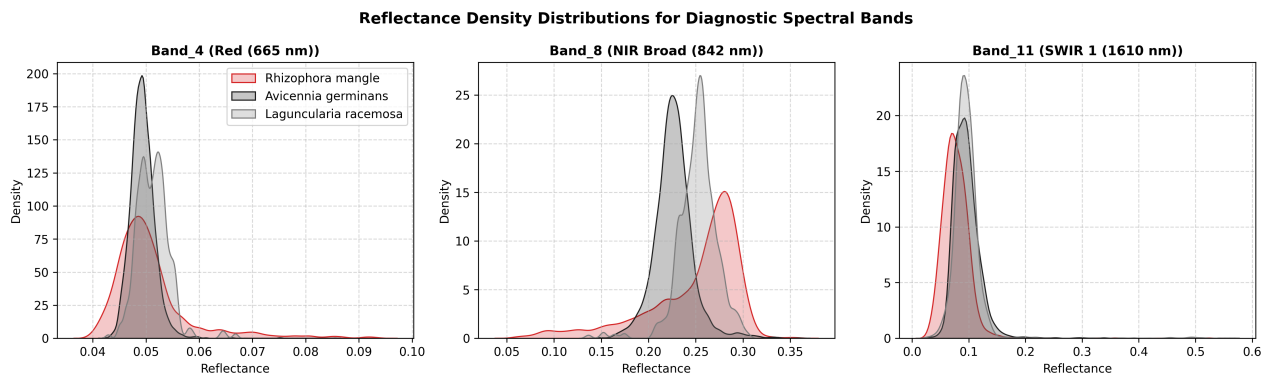


Figure 3. Reflectance density distributions for the three most diagnostic spectral bands: red (665 nm), broad near-infrared (842 nm) and first shortwave infrared (1610 nm).

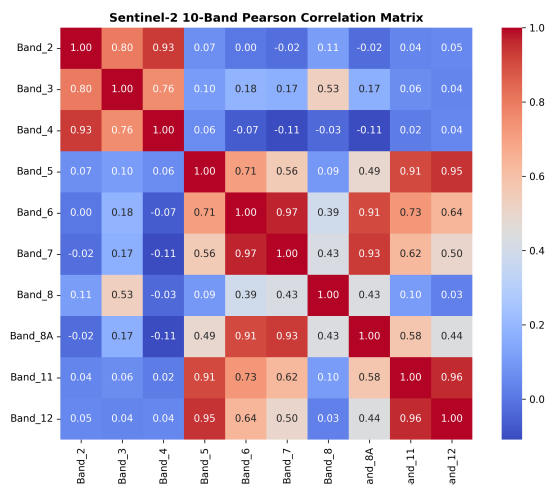


Figure 4. Pearson correlation matrix of the ten Sentinel-2 bands over the monospecific mangrove pixels, showing the visible and red-edge/shortwave infrared blocks.

Fig. 10 relates this pattern to model size, by plotting the gap against total parameter count on a logarithmic scale with depth encoded by colour and width by marker size. Across the 41-model grid, the gap tends to increase with parameter count rather than with depth alone: it remains near zero below roughly 10^3 parameters, increases through an intermediate range between 10^3 and 10^4 parameters, and is largest, at 4–5 percentage points, above 10^5 parameters, irrespective of the specific depth. The five-layer, 50-neuron model has approximately 10,800 parameters (Table 3) — more than an order of magnitude fewer than the 274,803 parameters of the four-layer, 300-neuron model — while still comprising five non-linear transformations of the input.¹ The association observed is consistent with the hypothesis that restricting layer width, rather than depth, constrains the model’s capacity to fit sample-specific noise in this dataset; it is a correlational pattern across ten architectures on a single held-out split, and, in the absence of multi-seed replication or an independently

¹The figure-generation script annotates this configuration as “10.9k parameters”, a rounding of a value close to but not identical to the 10,803 recorded in the underlying results table; [AUTHOR INPUT REQUIRED: reconcile the exact parameter count for MLP 5L–50N between `three_class_mangrove_results.csv` and the figure-generation script]. The discrepancy is on the order of 100 parameters out of roughly 10,800 and does not affect the qualitative pattern discussed here.

designed capacity-controlled experiment, it should be read as evidence consistent with the original authors’ qualitative interpretation rather than as an independent proof of a regularisation mechanism.

Fig. 11 and Fig. 12 decompose the errors by species. Under the five-layer, 50-neuron model the dominant confusions are reciprocal exchanges between *L. racemosa* and the other two species, while *A. germinans* attains the highest precision of the three (> 0.93), consistent with its comparatively tight reflectance envelope in Fig. 2 and its distinct shortwave infrared position. *L. racemosa* carries the lowest precision but a comparatively high recall, indicating a tendency of this model to over-assign that class in the held-out test partition — a pattern that runs in the same direction as the overestimation of *L. racemosa* that the original authors reported for the simpler spatial models, though the two observations are made on different evidence (tabular recall here, spatial maps in the original study) and are not formally linked. Under the four-layer, 300-neuron model all three species reach $F_1 > 0.91$ and the imbalance between precision and recall narrows. For the logistic baseline every species falls below $F_1 = 0.76$, consistent with linear hyperplanes being unable to separate the non-linear reflectance envelopes of the three species in this feature space.

3.4. Models test

The spatial evaluation reported in Section 3.3 of [Monterrubio-Martinez et al. \(2025\)](#) is thoroughly described in the original text but is not reproducible at raster level, because the Sentinel-2 scene GeoTIFFs, the 3.5 cm UAV orthophoto and the QGIS species polygons were not deposited. The status of this component is therefore **DATA-LIMITED**: the methodology is auditable, the outputs are not. No substitute imagery was obtained and no mask was synthesised.

What the audit preserves is a record of the original authors’ reported spatial findings, restated here as claims of the original study rather than as findings of this reproduction, together with the spectral evidence in the released data that bears on them. [Monterrubio-Martinez et al. \(2025\)](#) report that, for the binary models, accuracy exceeded 99 % for every architecture yet spatial quality varied substantially, with the logistic baseline producing extensive salt-and-pepper noise and false positives over water, and the four-layer, 300-neuron model delineating mangrove boundaries most cleanly. For the multiclass models, they report that the configuration with

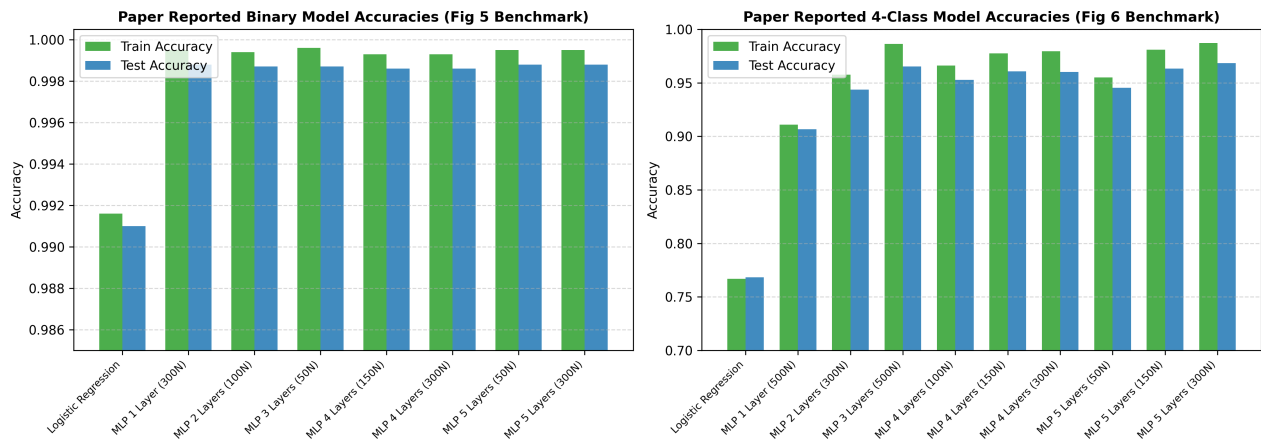


Figure 5. Accuracies reported by [Monterrubio-Martinez et al. \(2025\)](#) for the binary 60,000-pixel models (left, their Fig. 5) and the four-class 40,000-pixel models (right, their Fig. 6), transcribed here as reference benchmarks.

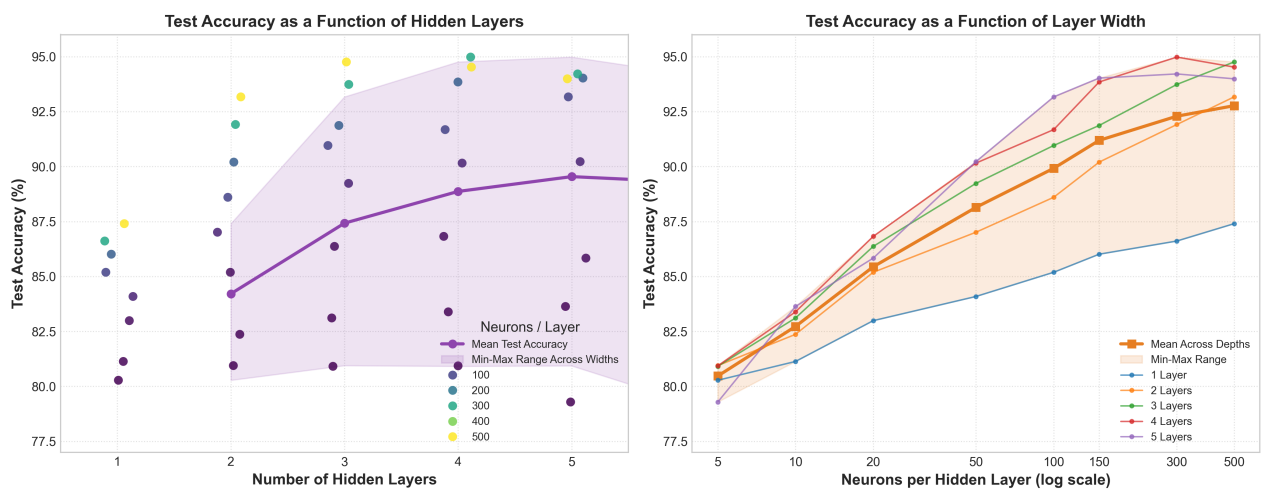


Figure 6. Test accuracy of the reproduced three-species models as a function of hidden-layer depth (left) and of layer width (right) across the complete 41-configuration grid.

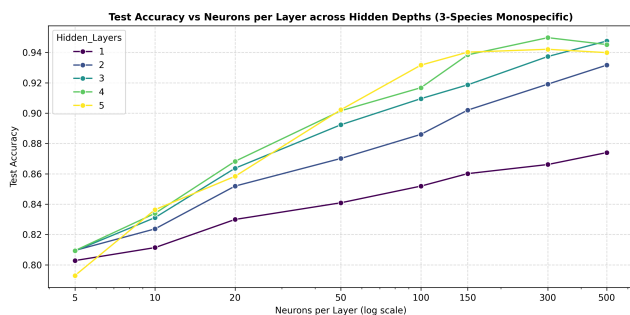


Figure 7. Test accuracy against neurons per layer for each hidden-layer depth on the three-species monospecific dataset.

the highest test accuracy suffered spectral confusion between *R. mangle* and surrounding water bodies, that the simpler models overestimated *L. racemosa*, and that the five-layer, 50-neuron model was judged the most reliable for real-world mapping despite its lower test accuracy. At Arroyo Moreno they report that the models produced plausible predictions without retraining on that site, with *R. mangle* again overestimated in permanently flooded zones while *A. germinans* and *L. racemosa* boundaries matched historical field surveys

([Lopez-Portillo et al., 2002](#)). None of these spatial outputs could be independently regenerated or inspected in this reproduction, and the description above should be read as a summary of the original text, not as an independently verified result.

Two features of the spectral evidence available in this reproduction are, at least qualitatively, compatible with two of these reported spatial observations, without constituting independent verification of either. The reported confusion of *R. mangle* with water is qualitatively consistent with Figs. 2 and 3: *R. mangle* has the lowest shortwave infrared reflectance of the three species in the released data and by far the widest dispersion, so its lower tail overlaps the reflectance region typically occupied by open and turbid water; the original authors attribute this to the higher moisture content of mangrove canopy. The reported tendency of simpler models to overestimate *L. racemosa* runs in the same direction as the per-class metrics of Fig. 12, where that species carries the lowest precision and the highest recall under the narrow five-layer model on the tabular three-class problem. These are consistency checks at the level of spectral plausibility, not confirmations of the spatial maps themselves, which remain outside the reach of this reproduction (Section 5).

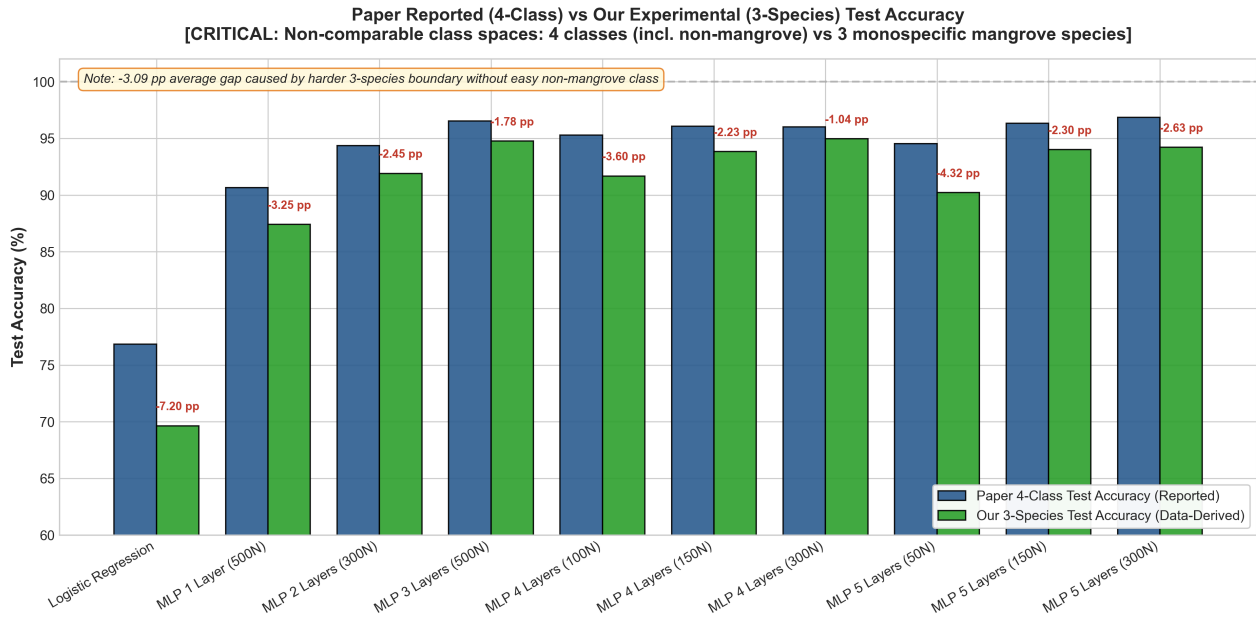


Figure 8. Not a direct performance comparison. Four-class test accuracies reported by [Monterrubio-Martinez et al. \(2025\)](#) against the three-species test accuracies obtained in this reproduction, with the percentage-point difference annotated. The two class spaces differ (three mangrove species here versus three species plus a non-mangrove background class in the original study), so the values are shown for audit transparency only and must not be read as a measure of relative model quality.

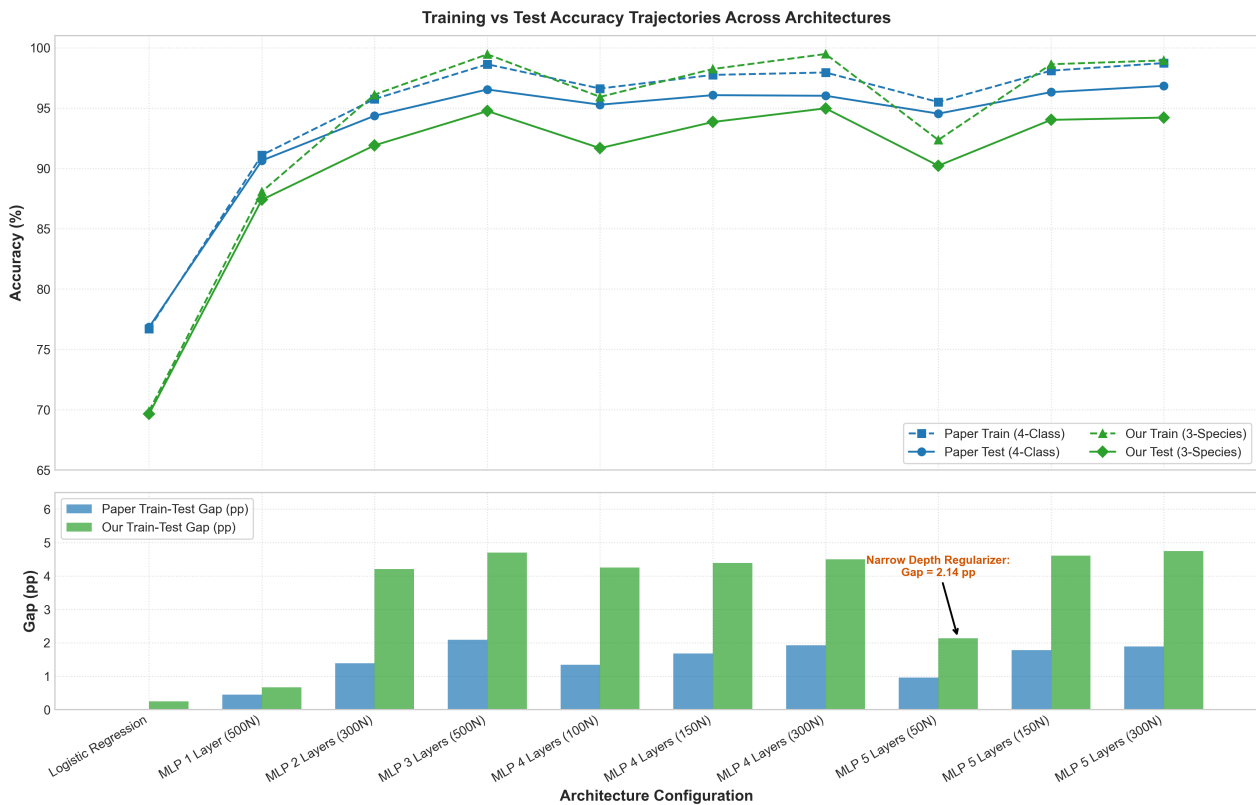


Figure 9. Training and test accuracy trajectories across architectures (top) and the corresponding train–test gap (bottom) for the reported four-class models and the reproduced three-species models. The narrow five-layer, 50-neuron configuration is annotated.

4. Reproducibility audit findings

Table 5 consolidates the file-by-file and claim-by-claim audit described in Section 2 into a single reproducibility-status matrix, using the same five-label evidence taxonomy introduced in Section 2.1. It is intended as a primary, citable output of

this study in its own right: a reader who wants to know, for a specific component of [Monterrubio-Martinez et al. \(2025\)](#), whether it can currently be independently verified, can consult this table directly rather than the discursive text.

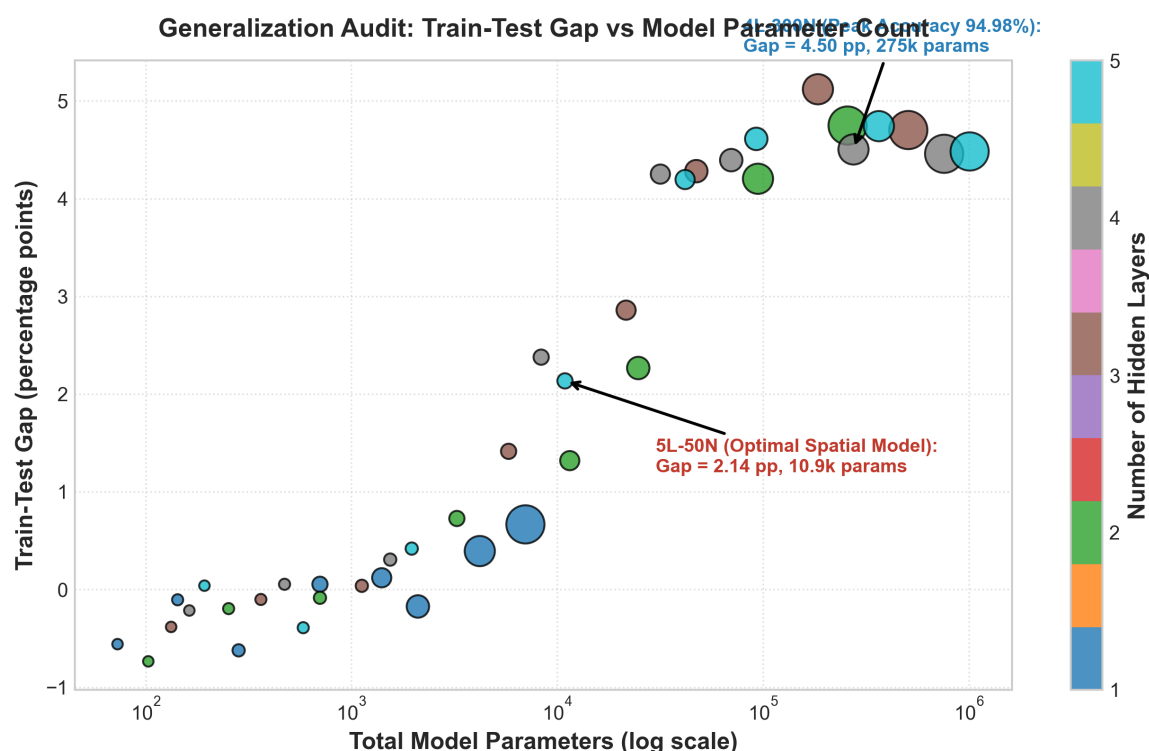


Figure 10. Train–test gap against total model parameter count (logarithmic scale) for the full 41-model grid, coloured by hidden-layer depth and sized by layer width.

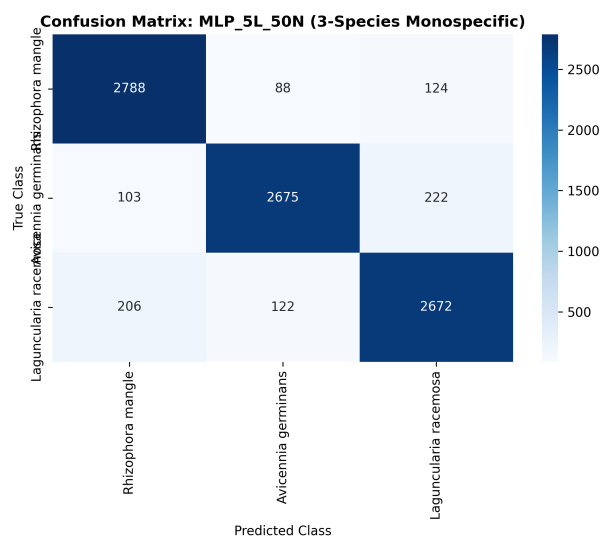


Figure 11. Confusion matrix of the five-layer, 50-neuron model on the 9,000-pixel test partition of the three-species monospecific dataset.

5. Threats to validity

This section makes explicit the limitations that constrain what can legitimately be claimed from the results in Section 3, organised into internal, statistical, external and reproducibility validity. None of these limitations is newly discovered here; they follow directly from the evidence audit in Sections 2 and 4 and are collected here so that a reader can assess, in one place, the conditions under which the reported numbers should be interpreted.

5.1. Internal validity

The original learning rate, batch size, epoch count, weight-initialisation scheme and random seeds are NOT AVAILABLE, and the documented IMPLEMENTATION ASSUMPTION values used here ($\alpha = 0.001$, batch size 256, 30 epochs, seed 42, PyTorch Kaiming-uniform initialisation) are not known to match the original implementation's. The original work cites a Keras/TensorFlow reference (Chollet, 2021), whose default Glorot-uniform initialisation differs from the PyTorch default used here, and framework-level differences in optimiser numerics, data-loader shuffling and floating-point behaviour are a further source of internal-validity uncertainty. The preprocessing order (whether standardisation was fit before or after the train/test split in the original study) is not stated in the text and is NOT AVAILABLE; this reproduction fits the scaler on the training partition only, which is stated as this study's own documented choice and not as a recovered fact about the original pipeline. The train/test split itself is performed at the pixel level with stratification by class label but without blocking by scene, acquisition date or spatial location, which creates the possibility, discussed under statistical validity below, that information from a given location leaks between partitions. All of these are threats to the internal validity of the numerical accuracies reported in Tables 3–4, in the sense that a different, equally defensible set of implementation choices could plausibly produce different accuracy values on the same data.

5.2. Statistical validity

The pixel-level observations that enter the ANOVA, the Tukey HSD tests and the classification accuracies are not statistically independent: multiple pixels are drawn from the same monospecific polygon, the same polygon is imaged repeatedly



Figure 12. Per-class precision, recall and F_1 -score for the logistic baseline, the five-layer 50-neuron model and the four-layer 300-neuron model on the three-species dataset.

Table 5. Reproducibility audit matrix. “Available” records whether the artefact or method detail is recoverable from the public deposit or the paper text; “Reproduced” records whether it was independently executed or verified in this study. Evidence status follows the taxonomy of Section 2.1.

Component	Required artefact / method	Avail.	Repro.	Evidence status	Limitation
Dataset inventory	File-level manifest of the KNB deposit	Yes	Yes	DATA-DERIVED	Manifest is limited to what was actually deposited (Sec. 2.2).
Public reflectance CSV	Released 191 MB CSV, 1,605,681 rows (...LaMancha.csv)	Yes	Yes	DATA-DERIVED	Contains only the three monospecific mangrove species, no background class.
Spectral bands (10)	B2–B12 surface reflectance, pre-resampled	Yes	Yes	PAPER-SPECIFIED / DATA-DERIVED	
Preprocessing (Standard-Scaler)	Fit-on-train, transform-train-and-test	Partial	Yes (this study)	PAPER-SPECIFIED / NOT AVAILABLE (fit order in original)	Original fit-before/after-split order not stated; this study’s own procedure is documented but is not evidence about the original implementation.
One-way ANOVA	Per-band F -test, 3 species	Yes	Yes	DATA-DERIVED	Large- n significance; no effect sizes computed (Sec. 3.1).
Tukey HSD	30 pairwise post-hoc contrasts	Yes	Yes	DATA-DERIVED	Same non-independence caveat as ANOVA.
Spectral signature curves	Mean $\pm$ SD reflectance per band	Yes	Yes	DATA-DERIVED	—
Band correlation analysis	Pearson correlation matrix	Yes	Yes	DATA-DERIVED	—
41-model MLP grid	Logistic + 1–5 layers $\times$ 8 widths	Yes (arch. spec.)	Yes, on 3-class data	PAPER-SPECIFIED (grid) / DATA-DERIVED (results)	Trained on 3-class, not on the original 4-class or binary label space.
Binary classification (60k px)	Mangrove vs non-mangrove pixels	No	No	DATA-LIMITED	Non-mangrove pixels not deposited; not attempted.
Four-class classification (40k px)	3 species + non-mangrove	No	No	DATA-LIMITED	Non-mangrove pixels not deposited; not attempted.
Non-mangrove samples	Rainforest, crops, water, urban, dunes, wetlands	No	No	DATA-LIMITED	Not deposited; not substituted or fabricated (Sec. 1).
UAV orthophoto	3.5 cm GeoTIFF, La Mancha	No	No	DATA-LIMITED	Not deposited.
QGIS species masks	Vector polygons, 3 species + mixed	No	No	DATA-LIMITED	Not deposited.
Sentinel-2 scene Geo-TIFFs	Full-scene rasters, La Mancha & Arroyo Moreno	No	No	DATA-LIMITED	Not deposited.
Spatial mapping (Figs. 7–9)	Pixel-wise prediction over scene rasters	No	No	DATA-LIMITED	Requires the scene GeoTIFFs above.
Spatial validation (Fig. 10)	Cross-site prediction at Arroyo Moreno	No	No	DATA-LIMITED	Requires the scene GeoTIFFs above.
Original hyperparameters	Learning rate, batch size, epochs	No	N/A	NOT AVAILABLE	Not stated in text; this study used documented IMPLEMENTATION ASSUMPTION values (Sec. 2.4).
Original random seed(s)	Sampling, split, initialisation seeds	No	N/A	NOT AVAILABLE	This study fixed seed 42 throughout; not the original authors’ value(s).
Cross-site / cross-scene split	Scene- or site-blocked train/test partition	No	No	NOT AVAILABLE	Neither study reports a spatially blocked evaluation; risk discussed in Section 5.

across 147 temporally correlated Sentinel-2 acquisitions of the same two sites, and neighbouring pixels within a scene are spatially autocorrelated. Classical ANOVA and the accuracy estimates reported here assume independent and identically distributed observations; that assumption is not satisfied by construction, so the very small p -values obtained ($p < 10^{-10}$ for every band) reflect both the underlying spectral separation and the large, non-independent sample size, and should not be interpreted as evidence of an unusually large effect.

For the same reason, the classification accuracies in Table 3 are best interpreted as performance under a stratified random pixel-level holdout on this specific multi-temporal pixel pool, rather than as an unbiased estimate of accuracy on genuinely independent, unseen locations or dates. No multi-seed replication was performed, so no variance or confidence interval is available for any reported accuracy or train–test gap [AUTHOR INPUT REQUIRED: multi-seed runs would allow this uncertainty to be quantified].

5.3. External validity

All results, in both the original study and this reproduction, come from a single study area (La Mancha, with an out-of-sample application at Arroyo Moreno reported only by the original authors and not independently verified here), a single sensor (Sentinel-2 L1C), a single acquisition period (2015–2021) and three mangrove species specific to the Gulf of Mexico coast of Veracruz. Neither the original study nor this reproduction establishes that the reported architectural or spectral-separability patterns transfer to other mangrove-forming species, other sensors, other geographic regions, or other combinations of monospecific versus mixed stands. Claims in this manuscript are accordingly restricted to the study area and species evaluated, and no general claim about mangrove species classification is intended.

5.4. Reproducibility validity

The non-mangrove background pixels, the subsampled 60,000-pixel binary and 40,000-pixel four-class training sets, the UAV orthophoto, the QGIS vector masks and the Sentinel-2 scene GeoTIFFs described in the original methodology are DATA-LIMITED: described in the text but absent from the public deposit (Table 5). As a direct consequence, the original binary classification experiment, the original four-class classification experiment, and the spatial mapping and cross-site validation of Figs. 7–10 cannot be independently reproduced from public evidence, and are not reproduced in this manuscript. The original specification is also incomplete on several points needed for exact reproduction (Section 5.1 above), which independently limits reproducibility even where the relevant data are available. These limitations are structural properties of the public record for this study, not shortcomings of the reproduction effort, and they define the boundary of what any independent audit of this study can currently establish.

6. Discussion

This study indicates that the statistical foundation of [Monterrubio-Martinez et al. \(2025\)](#) can be reproduced from the publicly deposited data, and that its principal architectural interpretation is not contradicted by an independently computed, complementary measure, while its headline accuracy figures remain outside what the public record supports. This is a scope statement about what has and has not been independently verified, not a ranking of the two studies; the distinction matters because the three components carry different evidential weight for a reader deciding how much to rely on the original method.

The exploratory result was reproduced from the released data. Every band separates the three species at $p < 10^{-10}$ and all thirty pairwise Tukey contrasts are significant, providing evidence consistent with the premise that Sentinel-2 reflectance carries species-level information in Veracruz mangroves, subject to the statistical-validity caveats of Section 5 regarding pixel non-independence and effect size. The practical consequence emphasised by the original authors, that multispectral data can be used directly to identify mangrove distribution without additional processing such as vegetation indices, is not contradicted by the 94.98 % three-species accuracy obtained here from raw standardised reflectance alone on a different (three-class, not four-class) class space.

The architectural interpretation is examined here through a different instrument than the one the original authors used. Their preference for the five-layer, 50-neuron model rested on visual inspection of predicted maps, which are not available for this reproduction. The train–test gap analysis in Section 3.3 offers a measurable, though indirect, complement to that judgement: at approximately 10,800 parameters (Table 3; see the parameter-count note in Section 3.3) the model shows a gap of 2.14 percentage points where models of comparable depth and greater width show gaps of 4.4 to 4.7 points. This pattern is consistent with, rather than proof of, the interpretation offered by the original authors that restricting neurons per layer may limit overfitting to spectral noise ([Hagan et al., 2014](#)); it is based on a single seed and ten of the forty-one configurations, and no controlled experiment isolating width from depth, sample size or training duration was performed. The pattern also illustrates a methodological point of wider relevance: in operational remote sensing, where the deliverable is a map rather than a confusion matrix, model selection on test accuracy alone carries risk, and capacity diagnostics of the kind computed here may be a useful, low-cost complement to spatial evaluation where the latter is unavailable ([Kumaraperumal et al., 2022](#); [Chowdhury, 2024](#)).

The headline accuracies of the original binary and four-class experiments could not be reproduced, and the reason is structural rather than a matter of implementation effort. The deposited file contains only monospecific mangrove pixels; the non-mangrove background class described in Section 2.1 of the original article, sampled from rainforest, crops, water, urban areas, dunes and other wetlands, was not released. Both published experiments depend on it — one of two classes in the binary task and one of four in the multiclass task — so neither could be rebuilt without either obtaining the missing data from the original authors or fabricating a substitute, both of which are outside the scope of this study. The −3.09 percentage point mean offset between the reported four-class accuracies and the three-species accuracies obtained here is, accordingly, attributable to a different class space, with an additional non-mangrove class and a different chance baseline (25 % against 33.3 %), and is shown in Table 4 and Fig. 8 for audit transparency rather than as a performance comparison. Stating that this reproduction performed worse, or better, than the original study would not be supported by the evidence available.

Several secondary sources of divergence between the two studies remain unresolved and are recorded as such rather than adjudicated. The learning rate, batch size, epoch count and random seeds are absent from the original text, and variation in these quantities can plausibly move test accuracy by more than a percentage point; the values adopted here are documented but are not known to match the original authors'. The framework differs, Keras-style Glorot-uniform initialisation against PyTorch Kaiming-uniform, which is a further obstacle to numerical agreement. The original article also does not state whether the 70/30 split was stratified or whether pixels were partitioned across the 147 acquisitions; as set out in Section 5, unblocked pixel-level splitting over a multi-temporal stack raises the possibility that the same ground location, imaged on different dates, appears in both partitions of either study, which would be expected to inflate pixel-level test accuracy relative to genuine out-of-scene gen-

eralisation. This is raised as an unverified structural risk that applies to the splitting procedure described in the original text and to the splitting procedure used in this reproduction alike, not as a finding that leakage occurred in either case, which the available information is insufficient to establish.

Two minor inconsistencies were identified in the original text during the audit and are recorded for completeness. The description in Section 3.3 attributes a training score of 0.9775 and a test score of 0.9607 to a one-hidden-layer model with 150 neurons, whereas the caption of Fig. 9 and the surrounding context identify this configuration as a four-hidden-layer model; the figure caption appears to be correct. In addition, all numerical results are presented through bar charts and spatial maps, with only selected models quoted in the text, so no complete tabular record of the 41-model grid exists in the published article. Both observations are offered as reproducibility notes rather than as errors of substance, and neither affects the study's conclusions.

7. Recommendations for reproducible ecological machine learning

The findings above motivate a small set of concrete recommendations, addressed primarily to future data deposition and evaluation practice in this subfield rather than to the original authors specifically.

Deposition completeness. The authors deposited a substantial and well-documented dataset, and it is precisely because the monospecific pixels were released that the statistical analysis and the architectural scaling behaviour could be examined at all. Extending the same deposit to the non-mangrove background table, to the two subsampled datasets actually used for training, and to the random seeds and optimiser settings would allow the published accuracies to be independently verified at comparatively low additional cost. Where scene GeoTIFFs are too large to archive routinely, depositing the scene identifiers and acquisition dates, together with the vector masks, would allow an independent party to rebuild the spatial test from the Copernicus archive.

Spatially or temporally blocked evaluation. The pixel-level-leakage risk described in Section 5 could be addressed at the design stage by partitioning pixels by scene or by spatial block rather than at random, and by reporting accuracy on a held-out acquisition or a held-out spatial region. Given the modifiable areal unit problem and its implications for mangrove extent estimates (Jelinski and Wu, 1996), an evaluation protocol that separates spatial from temporal generalisation would strengthen claims of transferability such as the Arroyo Moreno test reported by the original authors.

Multi-seed reporting. Neither the original study nor this reproduction reports results from more than a single run per configuration. Reporting mean and spread across several random seeds, particularly for the architectures singled out for their generalisation behaviour, would allow a reader to judge whether differences such as the 2.14-percentage-point train-test gap of the five-layer, 50-neuron model in this reproduction are stable or fall within run-to-run variability [AUTHOR INPUT REQUIRED: multi-seed results for this reproduction, if produced in a future revision].

Effect sizes alongside significance tests. Given the very large and non-independent pixel counts involved, reporting a

standardised effect size (for example, η^2 per band) alongside each ANOVA result would let readers separate statistical detectability from ecological or spectral importance [AUTHOR INPUT REQUIRED: effect-size computation, not performed in the present reproduction].

Context-aware architectures. The direction of future work indicated by the original authors — moving beyond pixel-wise perceptrons to segmentation architectures such as U-Net that incorporate spatial context alongside spectral response — is consistent with the error structure documented here. The residual confusions reported by the original authors and reflected in the tabular per-class metrics of this reproduction, the confusion of *R. mangle* with water in flooded zones and the overestimation of *L. racemosa*, are spatially organised phenomena that a classifier operating on one pixel at a time cannot, by construction, resolve. Reproducible evaluation of such context-aware models will depend on the deposition practices described above.

8. Conclusions

This study conducted an independent reproduction and reproducibility audit of Monterrubio-Martinez et al. (2025), applying a five-label evidence taxonomy to every element of the original methodology. The scope of this conclusion is limited to what the evidence assembled in Sections 3–5 supports; it is not a re-adjudication of the original study's own conclusions, which this work is not positioned to overturn.

What was reproduced. The exploratory and statistical analysis was reproduced from the publicly deposited data: one-way ANOVA on each of the ten Sentinel-2 bands and all thirty pairwise Tukey HSD contrasts return significant differences between *R. mangle*, *A. germinans* and *L. racemosa* ($p < 10^{-10}$ and $p < 0.001$ respectively), consistent with the original study's Fig. 4, subject to the pixel-non-independence caveat of Section 5. The 41-configuration architectural grid was re-implemented and trained on a balanced 30,000-pixel three-species dataset derived from the deposited file under a stratified pixel-level holdout, yielding test accuracies from 69.64 % for the logistic baseline to 94.98 % for the four-layer, 300-neuron model. This indicates that, under this experimental design, three co-occurring mangrove species are separable at better than 94 % from raw standardised reflectance without vegetation indices.

What was examined, but not proved. A train-test gap analysis found that the five-layer, 50-neuron configuration the original authors selected for spatial mapping shows a gap of 2.14 percentage points at approximately 10,800 parameters, smaller than the 4.4–4.7 point gaps of wider models of similar depth. This single-seed, ten-configuration observation is consistent with, and offers a quantitative complement to, the original authors' qualitative argument that narrower layers may limit overfitting; it does not, on its own, establish a causal regularisation mechanism, and it was not tested through a controlled capacity experiment or multiple seeds.

What was not, and could not be, reproduced. The published binary (60,000-pixel) and four-class (40,000-pixel) accuracies, and the full-scene spatial predictions of Figs. 7–10, are not reproducible from the public record, because the non-mangrove background pixels, the two subsampled training datasets, the UAV orthophoto, the QGIS vector masks and

the Sentinel-2 scene GeoTIFFs were not deposited (Table 5). These values are reported in this manuscript as reference benchmarks, shown alongside the three-species results for audit transparency only. The mean offset of -3.09 percentage points between them and the three-species results obtained here is attributable to a different class space and a different chance baseline, and does not constitute a comparison of model quality in either direction.

Limitations affecting interpretation. The undocumented original hyperparameters and random seeds, the framework difference between the likely Keras/TensorFlow original implementation and the PyTorch implementation used here, the unblocked pixel-level train/test split common to both studies, the single-seed design of this reproduction, and the restriction of both studies to one study area, one sensor and three species together bound what can be claimed from the results (Section 5).

Recommendation for future work. Readers seeking to build on either study should treat the reported accuracies as specific to the stratified random pixel-level holdout, the study area and the acquisition period evaluated, and should prioritise, where resources allow, spatially or temporally blocked evaluation splits, multi-seed replication, and deposition of the background-class data needed to close the binary and four-class reproduction gap identified here.

All reproduction code, notebooks and generated results supporting this audit are released publicly (see the Data and code availability statement at the end of this article) so that the audit itself remains open to independent scrutiny.

Description of author's responsibilities

Bala Sudalaimuthu: Conceptualisation, methodology, software, formal analysis, data curation, visualisation, writing — original draft, writing — review and editing.

Declaration of competing interest

The author declares no competing financial interests or personal relationships that could have appeared to influence the work reported in this paper. This study is an independent reproduction and was conducted without the involvement of the authors of the original article.

Acknowledgements

The author thanks Monterrubio-Martínez et al. for depositing the monospecific Sentinel-2 reflectance dataset in the Knowledge Network for Biocomplexity, without which none of the verification reported here would have been possible, and acknowledges the European Space Agency and the Copernicus programme for open access to Sentinel-2 data.

Data and code availability

Original dataset (publicly available). The dataset analysed by Monterrubio-Martínez et al. (2025) and audited in this study is openly available from the Knowledge Network for Biocomplexity (KNB) under package identifier doi:10.5063/F1FN14P1 (<https://knb.ecoinformatics.org/view/doi:10.5063/F1FN14P1>), titled “Sentinel-2 satellite reflectance of monospecific mangrove forests (*Rhizophora mangle*, *Avicennia germinans* and *Laguncularia race-*

mosa) at 10 m spatial resolution in La Mancha, Veracruz, Mexico from 2015–2021”, published 2024-11-07 under a Creative Commons Attribution 4.0 International (CC-BY-4.0) licence. It comprises the reflectance CSV (DataBase_Sentinel_2_Mangrove_LaMancha.csv, 191 MB, 1,605,681 rows) and an Ecological Metadata Language (EML v2.2.0) XML descriptor.

Original artefacts not publicly available. The non-mangrove background reflectance table, the subsampled 60,000-pixel binary and 40,000-pixel four-class training sets, the original authors' source code and model checkpoints, the 3.5 cm UAV orthophoto, the QGIS vector masks, and the Sentinel-2 scene GeoTIFFs for La Mancha and Arroyo Moreno were not found in the KNB deposit and are, to the knowledge of this study, not otherwise publicly released (Table 5).

Reproduction code, notebooks and results. All code, Jupyter notebooks, generated result tables and figure-generation scripts produced for this independent reproduction are released publicly at <https://github.com/notfawkes/Mangroves-Blue-Carbon.git>. The repository includes eight executed notebooks covering the dataset audit, the statistical reproduction, the MLP re-implementation and the paper-versus-reproduction comparison; the modular source code (src/config.py, src/data/loader.py, src/models/mlp.py, src/stats/anova_tukey.py, src/training/trainer.py); the generated result tables (results/models/three_class_mangrove_results.csv, results/architecture_comparison.csv, results/anova_results.csv, results/tukey_results.csv); the twelve figures reproduced in this manuscript (results/figures/); the discrepancy-audit documents underlying Sections 2, 4 and 5 (results/PAPER_VS_REPRODUCTION.md, SPATIAL_REPRODUCTION_STATUS.md); and a small unit-test suite (tests/test_reproduction.py).

Computational environment. The reproduction was executed in Python 3.9.6 with PyTorch 2.8.0 (Apple Silicon, Metal Performance Shaders backend), scikit-learn 1.6.1, SciPy 1.13.1, statsmodels 0.14.6, pandas 2.3.3 and NumPy 2.0.2, as against the paper's stated Python 3.8.13 and scikit-learn 1.2.1 with an implied Keras/TensorFlow backend (Chollet, 2021). Exact package versions are pinned in the repository's environment specification.

References

- Chollet, F., 2021. Deep Learning with Python, 2nd ed. Manning Publications.
- Chopade, M.R., Mahajan, S., Chaube, N., 2023. Assessment of land use, land cover change in the mangrove forest of Ghogha area, Gulf of Kambhat, Gujarat. Expert Syst. Appl. 212. <https://doi.org/10.1016/j.eswa.2022.118839>.
- Chowdhury, M.S., 2024. Comparison of accuracy and reliability of random forest, support vector machine, artificial neural network and maximum likelihood method in land use/cover classification of urban setting. Environ. Chall. 14, 100800. <https://doi.org/10.1016/j.envc.2023.100800>.
- FAO, 2023. The World's Mangroves 2000–2020. Food and Agriculture Organization of the United Nations, Rome.

- Hagan, M.T., Demuth, H.B., Beale, M.H., De Jesús, O., 2014. Neural Network Design, 2nd ed.
- Jelinski, D.E., Wu, J., 1996. The modifiable areal unit problem and implications for landscape ecology. *Landscape Ecol.* 11 (3), 129–140. <https://doi.org/10.1007/BF02447512>.
- Kumaraperumal, R., Pazhanivelan, S., Geethalakshmi, V., et al., 2022. Comparison of machine learning-based prediction of qualitative and quantitative digital soil-mapping approaches for eastern districts of Tamil Nadu, India. *Land* 11 (12). <https://doi.org/10.3390/land11122279>.
- López-Portillo, J., Martínez, M.L., Hesp, P., et al., 2002. Atlas de las costas de Veracruz: Manglares y Dunas. Universidad Veracruzana.
- Monterrubio-Martínez, E., Trujillo-Acatitla, R., Tuxpan-Vargas, J., Moreno-Casasola, P., 2025. An approach for accurate identification and monitoring of species in mangrove forests based on multi-source spectral data and deep learning. *Ecol. Inform.* 85, 102961. <https://doi.org/10.1016/j.ecoinf.2024.102961>.
- Moreno Casasola, P., Rojas Galaviz, J.L., Zárate Lomelí, D., et al., 2002. Diagnóstico de los manglares de Veracruz: distribución, vínculo con los recursos pesqueros y su problemática. *Madera y Bosques* 8, 61–88. <https://doi.org/10.21829/myb.2002.801292>.
- Pham, T.D., Yokoya, N., Bui, D.T., Yoshino, K., Friess, D.A., 2019. Remote sensing approaches for monitoring mangrove species, structure, and biomass: opportunities and challenges. *Remote Sens.* 11 (3), 1–24. <https://doi.org/10.3390/rs11030230>.